ReqCheqAI

Kondwani & Grant

Project Requirements

1. The user needs to be able to input their requirement.
2. The analysis process needs to be started by a button the user can click.
3. ReqCheq will generate a report that consists of:
    a. Functional & Non-Functional Detection
    b. Ambiguity Scoring
    c. Active Voice Detection
    d. Overall Requirement Score
4. **Scoring Requirements Added During Development

Project Implementation | Backend

Two Endpoints:

1) Analyze → Input & Output Validation
   - LLM prompt + Score computations + DB save + Returns Analysis
2) Feedback → Input Validation
   - LLM prompt + Returns Feedback

- spaCy: NLP python library
  - Atomicity → Counts verbs
  - Measurability → detects numerical descriptions(like_num function)
  - Complexity → counts clauses
- textstat: calculates statistics from text
  - Readability → flesch_reading_ease function()
- Overall Score: manual grading criteria

Project Implementation | Frontend

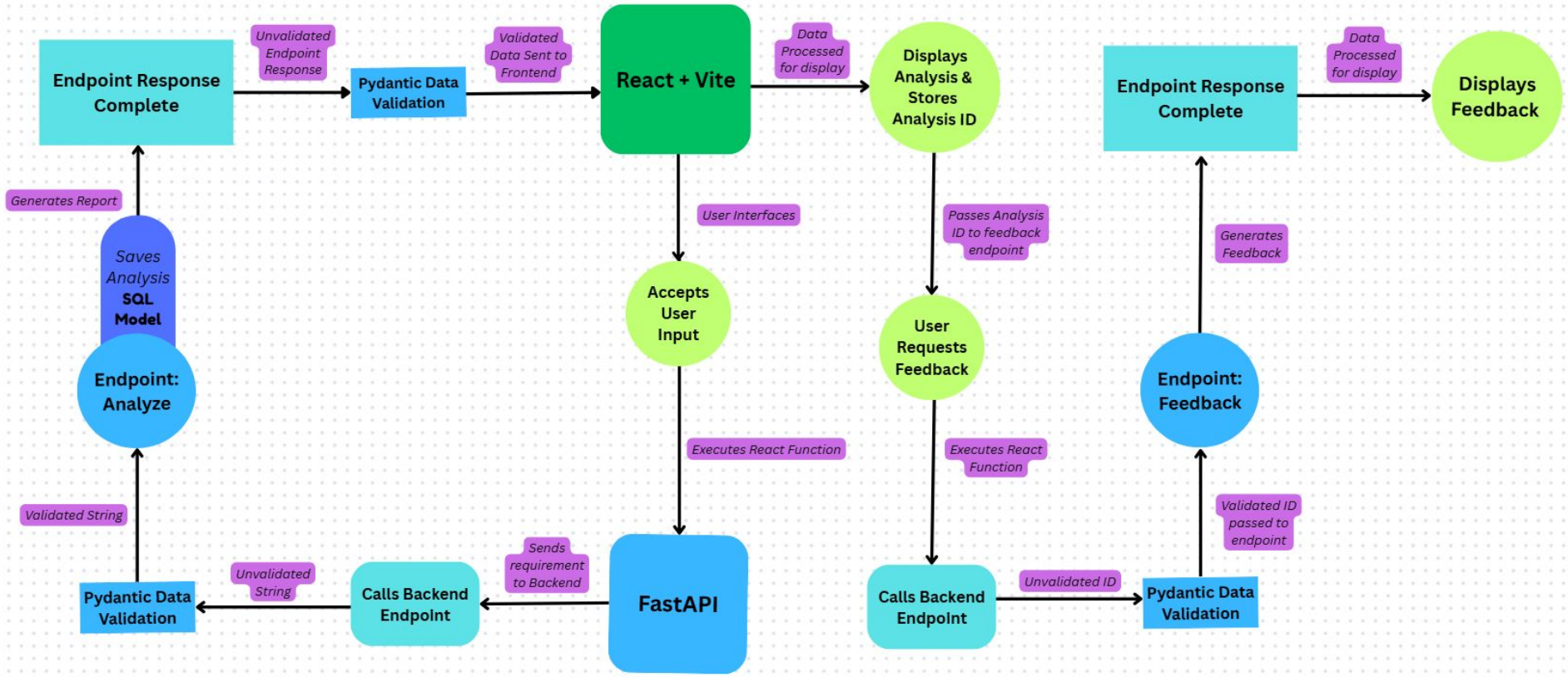
- Container for User Interaction
- Linking Components
  - Frontend → Backend → LLM API
- Styling Utilizing Modules

```
const [response, setResponse] = useState("")
    const [report, setReport] = useState("")
    const [atomicScore, setAtomicScore] = useState("")
    const [measureScore, setMeasureScore] = useState("")
    const [complexScore, setComplexScore] = useState("")
    const [readScore, setReadScore] = useState("")
    const [reqScore, setReqScore] = useState("")
    const [isLoading, setIsLoading] = useState(false)

    // Feedback Endpoint State Variables
    const [reqcheq_id, setreqcheq_id] = useState(null)
    const [feedback, setFeedback] = useState(null)
    const [isLoadingFeedback, setIsLoadingFeedback] = useState(false)

    async function submitHandle() {
        // For DevTools + Manual Testing, shows the button is actually registering a button click when pressed.
        console.log("Button clicked")
        if (!inputValue.trim()) return
        setIsLoading(true)
        setResponse("")
        setReport("")
        setAtomicScore("")
        setMeasureScore("")
        setMeasureScore("")
        setComplexScore("")
        setReadScore("")
        setReqScore("")
        // Feedback Endpoint States
        setreqcheq_id(null)
        setFeedback(null)
```

Project Design

- Frontend | React + Vite
  - Async React Functions
  - Returns JSX
- Backend | FastAPI
  - Pydantic
  - spaCy
  - Textstat
  - OpenAI API
- Database | SQL Model

Design: Data Flow



Testing

- Test Driven Development Cycles
  - Database Functionality → Saving Analysis Report
  - Score Computations → Atomicity, Measurability, Complexity, Readability
- Automated Tests:
  - Overall Score → Dummy Requirement Report
    - Simple Assert Equal
- Manual Testing:
  - LLM Responses → FastAPI docs
  - Frontend → Console, Terminal Error, Visual

Live Demo

AI in Development - For David :)

- Coding Assistant
  - Error Analysis & Solutions
- Design Recommendations
  - Textstat & spaCy for text processing
  - DB endpoint decorator removal
- Use in Endpoints
  - Analysis relies on LLM call
  - Feedback relies on LLM call
- Frontend / React Development
  - Used as a knowledge tool
  - Did not use it to just generate solutions